

Supplementary Materials for

Symmetric Mass Generation Transition and its Nonequilibrium Critical Dynamics in a Bilayer Honeycomb Lattice Model

First, we provide the details of the determinant quantum Monte Carlo (DQMC) method employed to calculate the observables in the ground state and those for the driven dynamics (Supplementary Note 1). Second, we present the structure factors and correlation-length ratios for charge density wave (CDW), spin density wave (SDW), and superconductivity (SC) orders. These results demonstrate the absence of phase transitions to CDW, SDW, and SC order (Supplementary Note 2). Third, we also show examples of how the single-particle gap is determined (Supplementary Note 3). Finally, we present the weighted nonlinear least-squares method for finite-size scaling analysis, which is used to fit the numerical data and obtain the critical point J_c and correlation-length exponent ν (Supplementary Note 4).

SUPPLEMENTARY NOTE 1: DETERMINANT QUANTUM MONTE CARLO

In this paper, we use DQMC to study the ground state properties and nonequilibrium processes in interacting fermion systems. The ground state expectation value of an observable $\hat{O}$ can be obtained by projecting a trial wave function $|\Psi_T\rangle$ along the imaginary time axis. When the projection time Θ is long enough, the trial initial state can be projected to the ground state, $|\Psi_0\rangle \propto \lim_{\Theta \rightarrow \infty} e^{-\Theta \hat{H}} |\Psi_T\rangle$. The expectation value of the ground state observables can be expressed as follows:

$$\langle \hat{O} \rangle = \frac{\langle \Psi_0 | \hat{O} | \Psi_0 \rangle}{\langle \Psi_0 | \Psi_0 \rangle} = \lim_{\Theta \rightarrow \infty} \frac{\langle \Psi_T | e^{-\Theta \hat{H}} \hat{O} e^{-\Theta \hat{H}} | \Psi_T \rangle}{\langle \Psi_T | e^{-2\Theta \hat{H}} | \Psi_T \rangle}. \quad (S1)$$

In this work, we set $2\Theta = 2L + 6$ to guarantee the convergence to the ground state.

To accurately implement the projection, the Trotter decomposition is used to discretize the imaginary time. The form of the Trotter decomposition is:

$$e^{-\Theta \hat{H}} = (e^{-\Delta\tau \hat{H}_I} e^{-\Delta\tau \hat{H}_t})^M, \quad (S2)$$

where $\Delta\tau = \frac{\Theta}{M}$ is set to 0.025, which is small enough to guarantee numerical convergence.

In addition, the Hubbard-Stratonovich (HS) transformation is required to decouple the fermionic interaction. To apply the HS transformation, the interaction term of the model is written as follows:

$$\hat{H}_I = J \sum_i \mathbf{S}_{i,1} \cdot \mathbf{S}_{i,2} = \sum_i \frac{J}{4} [(S_{i1}^z + S_{i2}^z)^2 + (S_{i1}^x + S_{i2}^x)^2 + (S_{i1}^y + S_{i2}^y)^2 - (S_{i1}^z - S_{i2}^z)^2 - (S_{i1}^x - S_{i2}^x)^2 - (S_{i1}^y - S_{i2}^y)^2]. \quad (S3)$$

In the simulation, for a single term in Eq. (S3), we use the discrete HS transformation [1]:

$$e^{\Delta\tau W O^2} = \sum_{l=\pm 1, \pm 2} \gamma(l) e^{\sqrt{\Delta\tau W} \eta(l) O} + O(\Delta\tau^4), \quad (S4)$$

where $W = J/4$ and O denotes the quadratic term in Eq. (S3), $\gamma(l)$ and $\eta(l)$ have the following values: $\eta(\pm 1) = \pm\sqrt{2(3 - \sqrt{6})}$, $\eta(\pm 2) = \pm\sqrt{2(3 + \sqrt{6})}$, $\gamma(\pm 1) = 1 + \sqrt{6}/3$, $\gamma(\pm 2) = 1 - \sqrt{6}/3$.

After the HS transformation, the interaction term is expressed as:

$$e^{-\Delta\tau \hat{H}_I} = \prod_{\sigma_1=x,y,z} \left[\sum_{l_1^{\sigma_1}=\pm 1, \pm 2} \gamma(l_1^{\sigma_1}) e^{\hat{h}_1^{\sigma_1}} \right] \times \prod_{\sigma_2=x,y,z} \left[\sum_{l_2^{\sigma_2}=\pm 1, \pm 2} \gamma(l_2^{\sigma_2}) e^{\hat{h}_2^{\sigma_2}} \right], \quad (S5)$$

where $\hat{h}_1^{\sigma_1} = \sqrt{\frac{\Delta\tau J}{4\eta}} (l_1^{\sigma_1}) (S_{i1}^{\sigma_1} - S_{i2}^{\sigma_1})$, $\hat{h}_2^{\sigma_2} = i\sqrt{\frac{\Delta\tau J}{4\eta}} (l_2^{\sigma_2}) (S_{i1}^{\sigma_2} + S_{i2}^{\sigma_2})$. Then, using DQMC, we can calculate the physical observables.

For the driven dynamics, we need to prepare an appropriate initial state $|\Psi_0\rangle$ and then linearly vary the interaction strength J with imaginary time. In this work, we prepare the DSM initial state with $J_0 = 0$ and drive the system to cross the critical point by changing J as $J = J_0 + R\tau$. The expectation value of the observable $\hat{O}$ is:

$$\langle \hat{O}(\tau) \rangle = \frac{\langle \Psi(\tau) | \hat{O} | \Psi(\tau) \rangle}{\langle \Psi(\tau) | \Psi(\tau) \rangle} \quad (S6)$$

$$= \frac{\langle \Psi_0 | e^{-\Theta \hat{H}} U^\dagger(\tau, 0) \hat{O} U(\tau, 0) e^{-\Theta \hat{H}} | \Psi_0 \rangle}{\langle \Psi(\tau) | \Psi(\tau) \rangle}, \quad (S7)$$

where $|\Psi(\tau)\rangle = U(\tau, 0)|\Psi_0\rangle$, with $U(\tau, 0) = \mathcal{T} e^{-\int_0^\tau d\tau' H(\tau')}$ and $\mathcal{T}$ represents the time-ordering operator.

SUPPLEMENTARY NOTE 2: THE RESULTS OF STRUCTURE FACTORS AND CORRELATION-LENGTH RATIOS

In the main text, we present the results for the correlation-length ratio of the EC order, based on the corresponding structure factor defined as $S_{EC} \equiv L^{-2d} \sum_{i,j} (-)^{i+j} \left\langle \left(c_{i\uparrow 1}^\dagger c_{i\uparrow 2} + c_{i\downarrow 1}^\dagger c_{i\downarrow 2} \right) \left(c_{j\uparrow 1}^\dagger c_{j\uparrow 2} + c_{j\downarrow 1}^\dagger c_{j\downarrow 2} \right) \right\rangle e^{i\mathbf{q} \cdot (\mathbf{r}_i - \mathbf{r}_j)}$, where i (j) is even (odd) for sublattice A (B) and $\mathbf{q} = 0$. Here, we supplement the results for the structure factors of other potential long-range orders and the corresponding correlation-length ratios. The structure factor is defined as follows:

$$S(\mathbf{q}) \equiv \frac{1}{L^{2d}} \sum_{i,j} \langle A_i A_j \rangle e^{i\mathbf{q} \cdot (\mathbf{r}_i - \mathbf{r}_j)}, \quad (S8)$$

where A_i (A_j) denotes the field operator and $\mathbf{q}$ is the momentum. We calculate the structure factors for CDW, SDW and SC orders at $\mathbf{q} = 0$ as shown in Fig. S1 (a)-(c). Their field operators are defined as [2]:

$$A_{\text{CDW}}(i) \equiv (-)^i \sum_l (c_{i\uparrow l}^\dagger c_{i\uparrow l} + c_{i\downarrow l}^\dagger c_{i\downarrow l}), \quad (S9)$$

$$A_{\text{SDW}}(i) \equiv (-)^i \sum_l (c_{i\uparrow l}^\dagger c_{i\uparrow l} - c_{i\downarrow l}^\dagger c_{i\downarrow l}), \quad (S10)$$

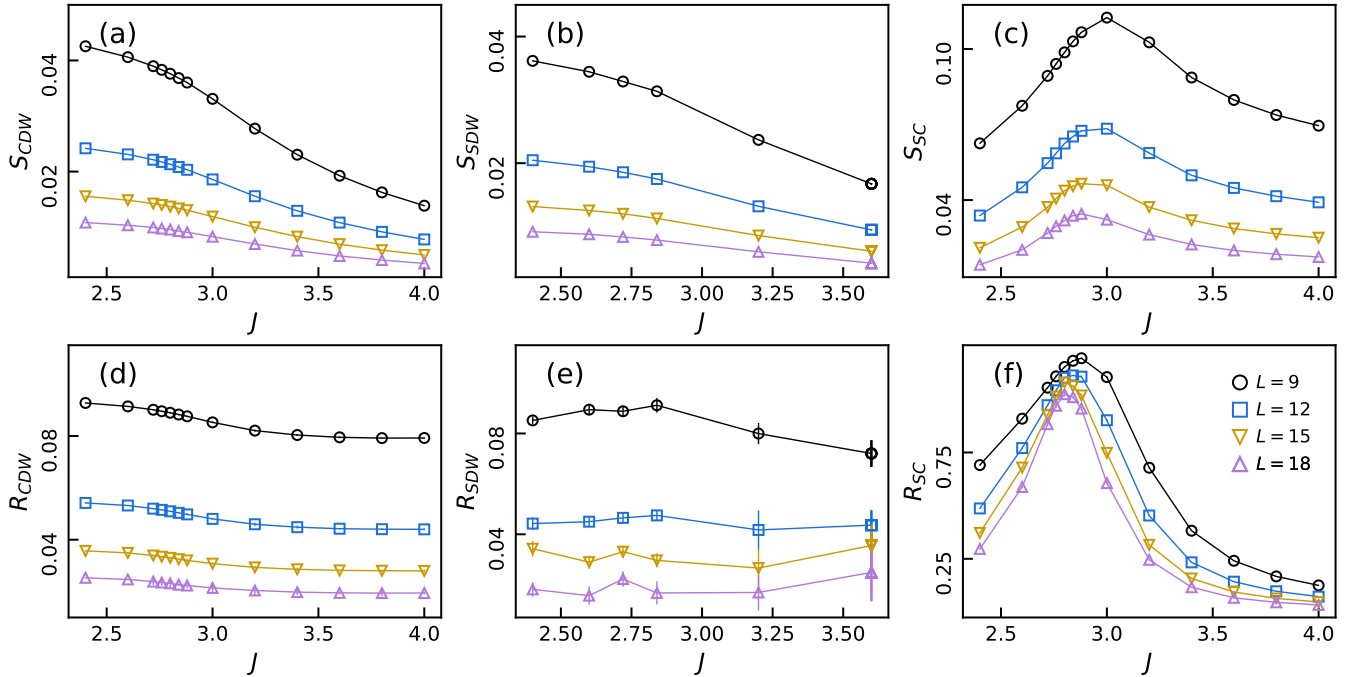


FIG. S1. **Exclusion of CDW, SDW, and SC orders.** The results of the structure factors (a)-(c) and correlation-length ratios (d)-(f) for CDW, SDW and SC as a function of J is shown for different system sizes. The lack of crossings in the curves of the correlation-length ratio suggests that no phase transition to long-range CDW, SDW, and SC order occurs.

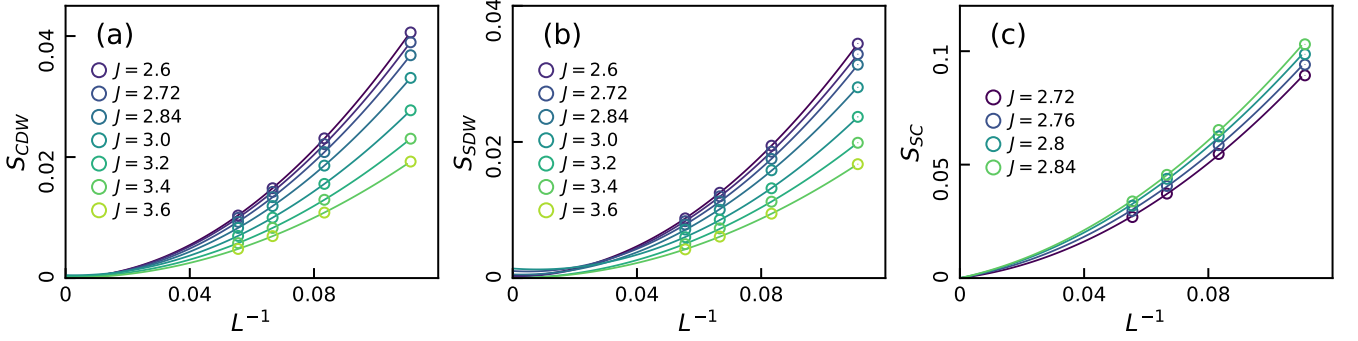


FIG. S2. **The extrapolated results of the structure factors for CDW, SDW, and SC.** The structure factors for $J > J_c$ are extrapolated as a function of L^{-1} , and the extrapolated values tend to zero within the error bar in the thermodynamic limit.

$$A_{SC}(i) \equiv c_{i\uparrow 1}^\dagger c_{i\downarrow 2}^\dagger - c_{i\downarrow 1}^\dagger c_{i\uparrow 2}^\dagger, \quad (S11)$$

where $l = 1, 2$ represents the index of the layer and i is even (odd) for sublattice A (B).

In addition, in Fig. S1 (d)-(f) we show the correlation-length ratio of different system sizes, which is defined as:

$$R(L) \equiv \frac{S(q, L)}{S(q + \Delta q, L)} - 1, \quad (S12)$$

where $\Delta q = (\frac{2\pi}{L}, \frac{2\pi}{L})$. The correlation-length ratios for CDW, SDW, and SC orders as a function of J are shown for different system sizes in Fig. S1 (d)-(f). The lack of crossings in these curves suggests that no phase transition to long-range CDW, SDW, and SC order occurs.

To further investigate whether spontaneous symmetry breaking occurs, we compute the structure factors of CDW, SDW, and SC orders for different system sizes in the region of $J > J_c$, and the extrapolations of these structure factors to the thermodynamic limit are shown in Fig. S2 (a)-(c). All extrapolated values tend to zero within the error bar. These results suggest the absence of long-range order corresponding to CDW, SDW, and SC during the phase transition.

SUPPLEMENTARY NOTE 3: SINGLE-PARTICLE GAP FROM IMAGINARY-TIME GREEN'S FUNCTION

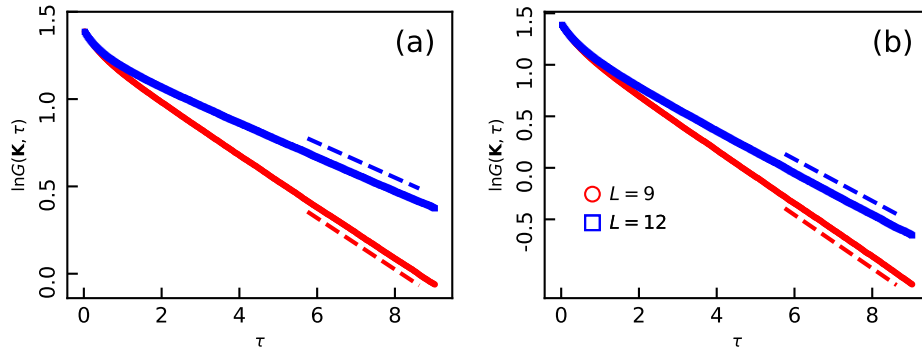


FIG. S3. **Determination of single-particle gap.** $\ln G(\mathbf{K}, \tau)$ as a function of imaginary time τ . (a) For $J = 2.4$ in the DSM phase, the single-particle gaps are determined to be $\Delta_{sp} = 0.146(1)$ for $L = 9$ and $0.099(1)$ for $L = 12$. The red dashed line and blue dashed line represent the fitted lines for $L = 9$ and $L = 12$, respectively. (b) For $J = 2.68$ in the SMG phase, the single-particle gaps are determined to be $\Delta_{sp} = 0.259(2)$ for $L = 9$ and $0.206(2)$ for $L = 12$. The red dashed line and blue dashed line represent the fitted lines for $L = 9$ and $L = 12$, respectively.

As discussed in the main text, the single-particle gap Δ_{sp} can be obtained from the imaginary-time Green's function $G(\mathbf{K}, \tau)$. For sufficiently large τ , $G(\mathbf{K}, \tau)$ satisfies $G(\mathbf{K}, \tau) \propto e^{-\Delta_{sp}(\mathbf{K})\tau}$.

Fig. S3 (a) and (b) show $\ln G(\mathbf{K}, \tau)$ as a function of imaginary time τ for different system sizes at $J = 2.4$ in the DSM phase and $J = 2.68$ in the SMG phase, respectively. By performing a linear fit to $\ln G(\mathbf{K}, \tau)$, the single-particle gaps Δ_{sp} are determined as $0.146(1)$ and $0.099(1)$ for $L = 9$ and $L = 12$ at $J = 2.4$, respectively. Following the same procedure, at $J = 2.68$, the corresponding values of Δ_{sp} for $L = 9$ and $L = 12$ are $0.259(2)$ and $0.206(2)$, respectively.

SUPPLEMENTARY NOTE 4: FINITE-SIZE SCALING ANALYSIS OF $L^z \Delta_{\text{sp}}$

In this work, we use weighted nonlinear least-squares fitting to determine the critical point J_c and the correlation-length exponent ν . Within the critical region, $L^z \Delta_{\text{sp}}$ satisfies the scaling relation

$$L^z \Delta_{\text{sp}} = \mathcal{F}((J - J_c)L^{1/\nu}). \quad (\text{S13})$$

To perform the finite-size scaling fit, we expand the scaling function $\mathcal{F}$ as a polynomial in its argument:

$$L^z \Delta_{\text{sp}} = \sum_{i=0}^n c_i ((J - J_c)L^{1/\nu})^i, \quad (\text{S14})$$

with polynomial order $n = 2$, where c_i are the fitting coefficients.

Using this polynomial form as the fitting model, we carry out weighted nonlinear least-squares fitting by minimizing the χ^2 residual,

$$\chi^2 = \sum_{\text{all data}} \frac{(L^z \Delta_{\text{sp}}^{\text{data}} - L^z \Delta_{\text{sp}}^{\text{fit}})^2}{\sigma^2(L^z \Delta_{\text{sp}}^{\text{data}})}, \quad (\text{S15})$$

where $L^z \Delta_{\text{sp}}^{\text{data}}$ denotes the DQMC data and $L^z \Delta_{\text{sp}}^{\text{fit}}$ is the corresponding fit value. $\sigma(L^z \Delta_{\text{sp}}^{\text{data}})$ denotes the standard error of the mean of the corresponding DQMC data. We use system sizes $L = 9, 12, 15, 18$ for the fitting, over the range $2.48 < J < 2.58$. To assess the quality of the fit, we evaluate the reduced chi-square, defined as $\chi_r^2 \equiv \chi^2/N_{\text{df}}$, where $N_{\text{df}} = N_{\text{data}} - N_{\text{params}}$ is the number of degrees of freedom. Here N_{data} is the total number of data points, and N_{params} is the number of fitting parameters. We obtain $\chi_r^2 \approx 1.26$ confirming that the model describes the data well within the statistical uncertainties.

The quoted uncertainties for $J_c = 2.584(4)$ and $\nu = 1.025(59)$ are the standard deviations of the fitted parameters, obtained from the covariance matrix of the nonlinear least-squares fit. The covariance matrix $\mathbf{C}$ is given by

$$\mathbf{C} = (\mathcal{J}^T \mathbf{W} \mathcal{J})^{-1}, \quad (\text{S16})$$

where $\mathcal{J}$ is the Jacobian matrix evaluated at the optimal parameters, and $\mathbf{W}$ is the diagonal weight matrix with entries $W_{kk} = 1/\sigma_k^2$, where k indexes the data points and σ_k is the corresponding statistical error. The standard deviation of each fitted parameter p_i is given by the square root of the corresponding diagonal entry of $\mathbf{C}$:

$$\sigma_{p_i} = \sqrt{C_{ii}}, \quad (\text{S17})$$

where C_{ii} is the corresponding diagonal entry of $\mathbf{C}$.

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- [1] F. Assaad and H. Evertz, “World-line and determinantal quantum monte carlo methods for spins, phonons and electrons,” in *Computational Many-Particle Physics*, edited by H. Fehske, R. Schneider, and A. Weiße (Springer Berlin Heidelberg, Berlin, Heidelberg, 2008) pp. 277–356.
- [2] W.-X. Chang, S. Guo, Y.-Z. You, and Z.-X. Li, [arXiv:2311.09970](https://arxiv.org/abs/2311.09970) (2023).